

Task

Deploy a SIEM of your choice (Elastic / Splunk / Microsoft Sentinel).

Set up a Windows VM with an agent/forwarder (Elastic Agent, Splunk UF, or AMA) shipping Windows Security Event Log to your SIEM.

Create and test 3 detection rules:

- Brute force — multiple failed logons for the same account (Event ID 4625, threshold-based)
- Local/domain user creation (Event ID 4720)
- User added to a privileged group, e.g. Administrators (Event IDs 4728 / 4732 / 4756)

Each rule must be properly documented, production-style — this is part of the grade: a clear name and description (what it detects and why it matters), severity, MITRE ATT&CK mapping (tactic + technique ID), and references.

Trigger each rule yourself on the VM (failed logons, net user /add, net localgroup administrators <user> /add).

SYSTEM ARCHITECTURE

The system consists of two main components that interact over the network:

Docker machine (Splunk server)

A Splunk Enterprise 9.1.2 container was deployed on the Docker machine, functioning as a centralized log collector and analyzer. Splunk listens on three ports: port 8000 for the web interface, port 8089 for management and coordination, and port 9997 for receiving logs from the Splunk Universal Forwarder installed on the Windows machine.

Windows VM

The Splunk Universal Forwarder was installed on the Windows VM. It automatically collects the Windows Security Event Log and sends it to the Splunk server on the Docker machine. The forwarder is configured to monitor three critical operating system logs: Security, Application, and System.

SPLUNK DEPLOYMENT AND CONFIGURATION

The first step was defining the network architecture. The IP address of the Docker machine was set to 192.168.0.252. This value is critically important, since the Windows VM uses this IP address to connect to Splunk.

To deploy Splunk in Docker, a docker-compose.yml file was created with the following configuration. This file allows Splunk to be started with a single command and ensures that the container automatically restarts on failure.

```

docker-compose.yml
1  version: '3.8'
2
3  services:
4    splunk:
5      image: splunk/splunk:${SPLUNK_VERSION}
6      container_name: splunk
7      environment:
8        - SPLUNK_START_ARGS=--accept-license
9        - SPLUNK_PASSWORD=${SPLUNK_PASSWORD}
10       - TZ=${TZ}
11      ports:
12        - "${SPLUNK_WEB_PORT}:8000" # Web UI
13        - "${SPLUNK_MGMT_PORT}:8089" # Management API
14        - "${SPLUNK_RECEIVING_PORT}:9997" # Forwarder Receiving Port
15      volumes:
16        - splunk_data:/opt/splunk/var
17        - splunk_etc:/opt/splunk/etc
18      restart: unless-stopped
19      healthcheck:
20        test: ["CMD", "curl", "-f", "http://localhost:8000/en-US/app/"]
21        interval: 30s
22        timeout: 10s
23        retries: 5
24
25      volumes:
26        splunk_data:
27          driver: local
28        splunk_etc:
29          driver: local

```

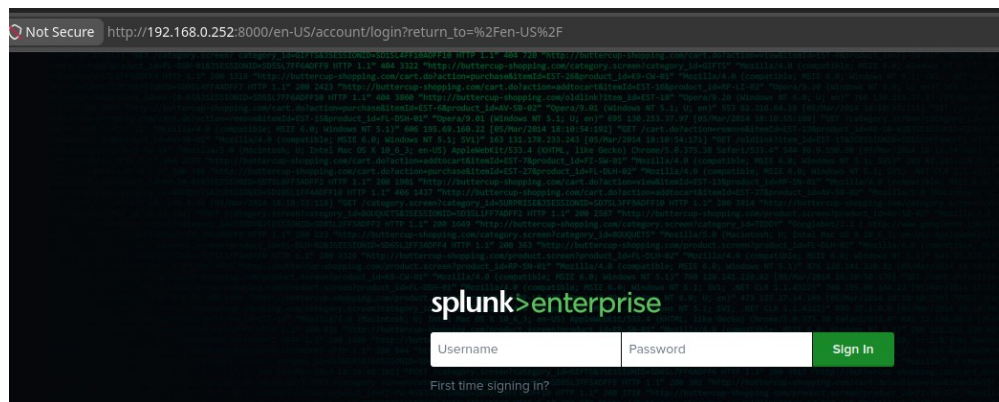
Splunk was started using the command `docker compose up`

```

* splunk-stem docker compose ps
WARN[0000] /home/mebius/splunk-stem/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME          IMAGE          COMMAND                  SERVICE    CREATED        STATUS               PORTS
splunk        splunk/splunk:9.1.2  "/sbin/entrypoint.sh..."  splunk     3 hours ago    Up 3 hours (healthy)  8065/tcp, 0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp, 8088/tcp, 8191/tcp, 0.0.0.0:8089->8089/tcp, [::]:8089->8089/tcp, 0.0.0.0:9997->9997/tcp, [::]:9997->9997/tcp, 9887/tcp

```

After the container was started, the accessibility of the Splunk web interface was verified.



To collect telemetry on the Windows VM, security event auditing (auditpol) was enabled and the Splunk Universal Forwarder was installed, which sends WinEventLog://Security logs to port 9997 of the SIEM server. The agent's operation was verified by restarting and checking the status of the SplunkForwarder service.

```
PS C:\Windows\system32> Get-Service SplunkForwarder

Status      Name                DisplayName
-----
Running     SplunkForwarder     SplunkForwarder
```

DETECTION RULE DEVELOPMENT

1. Brute force detection (Event ID 4625)

Brute force is an automated password-guessing attack, during which Windows logs every failed logon attempt as Event ID 4625. The detection logic is simple: instead of reacting to a single failure, all 4625 events are grouped by username and counted.

MITRE ATT&CK classification

- **Tactic:** Credential Access
- **Technique ID:** T1110 - Brute Force
- **Sub-technique ID:** T1110.001 - Password Guessing

The rule was created with the following query:

```
index=main EventCode=4625
| eval username=coalesce(TargetUserName, Account_Name, user, "Unknown")
| stats count as FailureCount by username, host
| where FailureCount >= 5
```

✓ 19 events (7/30/26 7:00:00.000 PM to 7/31/26 7:11:33.000 PM) No Event Sampling ▼

2. New user creation detection (Event ID 4720)

Creating a local account is often used by attackers to establish persistence. Windows logs every user creation event as Event ID 4720.

MITRE ATT&CK classification

- **Tactic:** Persistence
- **Technique ID:** T1136 - Create Account
- **Sub-technique ID:** T1136.001 - Local Account

The rule was created with the following query:

```
index=main EventCode=4720
| eval new_user=coalesce(TargetUserName, Account_Name, user, "New Account")
| table _time, host, new_user
```

✓ 3 events (7/30/26 7:00:00.000 PM to 7/31/26 7:15:02.000 PM) No Event Sampling ▼

Events Patterns **Statistics (3)** Visualization

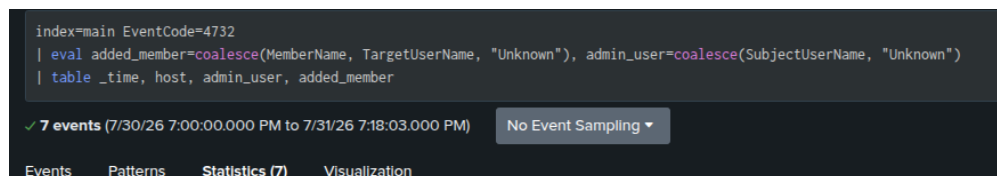
3. Detection of a user added to the Administrators group (Event ID 4732)

Granting a user administrator rights is the most critical event, since it represents privilege escalation. Windows logs the addition of users to any local group under Event ID 4732.

MITRE ATT&CK classification

- **Tactic:** Privilege Escalation
- **Technique ID:** T1548 - Abuse Elevation Control Mechanism
- **Sub-technique ID:** T1548.002 - Bypass User Account Control

The rule was created with the following query:



Saved alerts:

#	Title	Actions	Owner	App	Sharing	Status
1	Brute Force Detection - Multiple Failed Logins	Open in Search Edit	admin	search	App	Enabled
2	Local User Account Created	Open in Search Edit	admin	search	App	Enabled
3	User Added to Administrators Group	Open in Search Edit	admin	search	App	Enabled

DETECTION RULE TESTING

1. Brute Force detection

To test Brute Force detection, a password-guessing attack was simulated: logon attempts with an incorrect password for a single account were performed 6 times in a row.

```
PS C:\Windows\system32> $targetUser = "TestUser"
>> for ($i = 1; $i -le 6; $i++) {
>>   Write-Host "Attempt $i at $(Get-Date -Format 'yyyy-MM-dd HH:mm:ss')"
```

```
net use * \\localhost\c$ /user:$targetUser WrongPassword123 /persistent:no 2>$null
Start-Sleep -Seconds 10
}>>
```

```
Attempt 1 at 2026-07-31 19:54:14
Attempt 2 at 2026-07-31 19:54:28
Attempt 3 at 2026-07-31 19:54:42
Attempt 4 at 2026-07-31 19:54:56
Attempt 5 at 2026-07-31 19:55:11
Attempt 6 at 2026-07-31 19:55:25
PS C:\Windows\system32>
```

Alert verification in Splunk

Brute Force Detection - Multiple Failed Logins

Detects 5+ failed login attempts (Event ID 4625) indicative of brute force.

Enabled: Yes, [Disable](#)

App: search

Permissions: Shared In App. Owned by admin. [Edit](#)

Modified: Jul 31, 2026 4:44:05 PM

Alert Type: Real-time [Edit](#)

Trigger Condition: Per-Result [Edit](#)

Actions: 1 Action [Edit](#)

Add to Triggered Alerts

Trigger History

20 per page

	TriggerTime	Actions
1	2026-07-31 19:55:17 EEST	View Results

New Search

Save As Create Table View Close

Index=main EventCode=4625

| eval username=coalesce(TargetUserName, Account_Name, user, "unknown")

| stats count as FailureCount by username, host

| where FailureCount >= 5

Before date time

2 events (1/170 3:00:00.000 AM to 7/31/26 7:55:17.476 PM)

No Event Sampling

Job

Fast Mode

Events Patterns Statistics (2) Visualization

20 Per Page

Format Preview

username	host	FailureCount
TestUser	DESKTOP-9NDV3VK	6

The alert triggered successfully, detecting 6 failed logons for user TestUser. The threshold value (5+) was exceeded, which caused the alert to be generated. This result proves that the Brute Force detection rule functions correctly.

2. User Creation detection

To test User Creation detection, a new local user was created on the Windows VM.

```
Created user: AttackerAccount_2100869078
PS C:\Windows\system32> $newUser = "Attacker"
>> $pass = ConvertTo-SecureString "TempPass123!" -AsPlainText -Force
>> New-LocalUser -Name $newUser -Password $pass -FullName "Suspicious Account"
>>

Name      Enabled Description
-----
Attacker True

PS C:\Windows\system32>
```

Alert verification in Splunk

Local User Account Created

Detects creation of a new local user account (Event ID 4720)

Enabled: Yes, [Disable](#)

App: search

Permissions: Shared In App. Owned by admin. [Edit](#)

Modified: Jul 31, 2026 4:47:17 PM

Alert Type: Real-time [Edit](#)

Trigger Condition: Per-Result [Edit](#)

Actions: 1 Action [Edit](#)

Add to Triggered Alerts

Trigger History

20 per page

	TriggerTime	Actions
1	2026-07-31 20:00:29 EEST	View Results

New Search

Save As Create Table View Close

Index=main EventCode=4720

| 1 event (1/170 3:00:00.000 AM to 7/31/26 8:05:29.777 PM)

No Event Sampling

Job

Fast Mode

Before date time

Events Patterns Statistics Visualization

20 Per Page

Format

Time	Event
7/31/26 8:05:28.495 PM	LogName=Security EventCode=4720 EventType=8 ComputerName=DESKTOP-9NDV3VK Source=ntls host = DESKTOP-9NDV3VK source = WinEventLog\Security sourcetype = WinEventLog:Security

```
recordNumber=2539
Keywords=Audit Success
TaskCategory=User Account Management
OpCode=Info
Message=A user account was created.

Subject:
  Security ID:      S-1-5-21-654244614-1562676850-2720905111-1000
  Account Name:     Meblus
  Account Domain:   DESKTOP-0HVD3VK
  Logon ID:         0x351830

New Account:
  Security ID:      S-1-5-21-654244614-1562676850-2720905111-1002
  Account Name:     Attacker
  Account Domain:   DESKTOP-0HVD3VK

Attributes:
  SAM Account Name: Attacker
  Display Name:     <value not set>
  User Principal Name: -
  Home Directory:   <value not set>
  Home Drive:       <value not set>
  Script Path:      <value not set>
  Profile Path:     <value not set>
  User Workstations: <value not set>
  Password Last Set: <never>
  Account Expires:  <never>
  Primary Group ID: 513
  Allowed To Delegate To: -
  Old UAC Value:    0x0
  New UAC Value:    0x15
  User Account Control:
    Account Disabled
    'Password Not Required' - Enabled
    'Normal Account' - Enabled
  User Parameters:  <value not set>
  SID History:      -
  Logon Hours:      All

Additional Information:
  Privileges: -
Collapse
host = DESKTOP-0HVD3VK | source = WinEventLog:Security | sourcetype = WinEventLog:Security
```

The alert successfully detected Event ID 4720 and triggered on the creation of a new user. The results clearly show the details: the name of the newly created user, the creation time, and the host on which it occurred. This proves that the User Creation detection rule functions correctly.

3. Privilege Escalation detection

To test Privilege Escalation detection, a user was added to the Administrators group.

```
PS C:\Windows\system32> Add-LocalGroupMember -Group "Administrators" -Member "Attacker"
>> Write-Host "Added TestUser to Administrators group"
Added TestUser to Administrators group
PS C:\Windows\system32>
```

Alert verification in Splunk

User Added to Administrators Group

User added to Administrators group (Event ID 4732). Critical. Indicates privilege escalation or account compromise.

Enabled: Yes [Disable](#)

App: search

Permissions: Shared by App. Owned by admin. [Edit](#)

Modified: Jul 31, 2026 4:52:57 PM

Alert Type: Real-time [Edit](#)

Trigger Condition: [Per-Result](#) [Edit](#)

Actions: 1 Action [Edit](#)

Add to Triggered Alerts

Trigger History

20 per page

	TriggerTime	Actions
1	2026-07-31 20:09:02 EEST	View Results

```
> 7/31/26      07/31/2026 08:09:01.563 PM
8:09:01.563 PM LogName=Security
                EventCode=4732
                EventType=0
                ComputerName=DESKTOP-0HNV3VK
                SourceName=Microsoft Windows security auditing.
                Type=Information
                RecordNumber=2345
                Keywords=Audit Success
                TaskCategory=Security Group Management
                OpCode=Info
                Message=A member was added to a security-enabled local group.

                Subject:
                    Security ID:      S-1-5-21-654244614-1562676850-2720905111-1000
                    Account Name:      Meblus
                    Account Domain:     DESKTOP-0HNV3VK
                    Logon ID:           0x351830

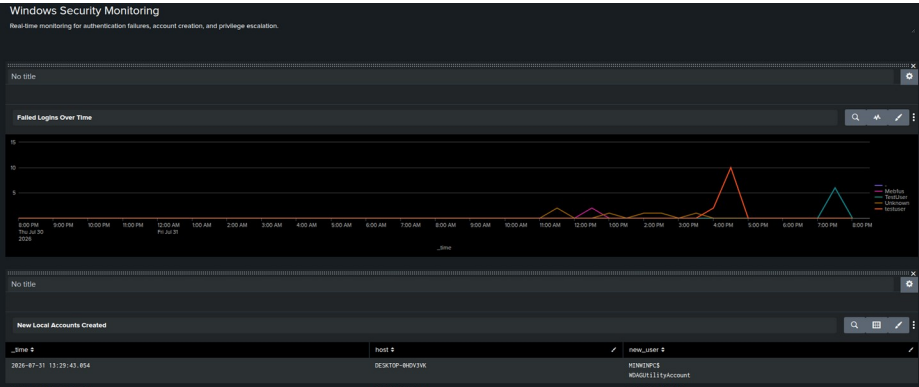
                Member:
                    Security ID:      S-1-5-21-654244614-1562676850-2720905111-1002
                    Account Name:      -

                Group:
                    Security ID:      S-1-5-32-544
                    Group Name:        Administrators
                    Group Domain:      BuiltIn

                Additional Information:
                    Privileges:        -
                Collapse
host = DESKTOP-0HNV3VK | source = WinEventLog:Security | sourcetype = WinEventLog:Security
```

The alert triggered successfully, detecting the addition of user TestUser to the Administrators group via Event ID 4732.

To ensure easy real-time monitoring of security events, a dedicated Dashboard was created in Splunk.



New Local Accounts Created		
_time	host	new_user
2026-07-31 13:29:43.854	DESKTOP-0HNV3VK	WIN-SSIS0NC1T1AS
2026-07-31 11:31:19.375	DESKTOP-0HNV3VK	Meblus
2026-07-31 13:19:03.253	DESKTOP-0HNV3VK	Meblus
2026-07-31 11:31:19.375	DESKTOP-0HNV3VK	TestUser
2026-07-31 20:05:28.495	DESKTOP-0HNV3VK	Meblus
2026-07-31 20:05:28.495	DESKTOP-0HNV3VK	Attackor

Privilege Escalation			
_time	host	admin_user	added_member
2026-07-31 13:29:58.021	DESKTOP-0HNV3VK	Unknown	Unknown
2026-07-31 13:29:58.020	DESKTOP-0HNV3VK	Unknown	Unknown
2026-07-31 13:29:43.853	DESKTOP-0HNV3VK	Unknown	Unknown
2026-07-31 11:31:19.398	DESKTOP-0HNV3VK	Unknown	Unknown
2026-07-31 11:31:19.378	DESKTOP-0HNV3VK	Unknown	Unknown
2026-07-31 16:48:48.524	DESKTOP-0HNV3VK	Unknown	Unknown
2026-07-31 15:56:28.035	DESKTOP-0HNV3VK	Unknown	Unknown
2026-07-31 20:05:01.563	DESKTOP-0HNV3VK	Unknown	Unknown

The deployment of the SIEM system on Splunk Enterprise was completed successfully. The system consists of a Splunk server in Docker, which receives logs from the Windows VM via the Splunk Universal Forwarder, and three

detection rules that detect Brute Force attacks, new user creation, and privilege additions to the Administrators group.

All three rules were tested and triggered correctly. The system is ready for continuous monitoring and detection of security threats on the Windows VM. Each rule is classified according to the MITRE ATT&CK framework, which allows the nature of detected threats to be correctly understood and responded to accordingly.

REFERENCES AND RESOURCES

- **MITRE ATT&CK Framework:** <https://attack.mitre.org/>
- **Windows Event Log Reference:** <https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/>
- **Splunk Documentation:** <https://docs.splunk.com/Documentation/Splunk>
- **Splunk Search Processing Language (SPL):** <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/>